Muhammad Saad Shabir

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EDUCATION

Carleton University **Ottawa, ON**
Bachelor of Information Technology - Network Technology *Expected Apr 2028*

SKILLS

Languages: Go, Rust, Python, C/C++, SQL, JavaScript/TypeScript, Bash
Tools & Technologies: Docker, Kubernetes, AWS, Linux, eBPF, Prometheus, Grafana, Wireshark, React.js, Node.js
Networking & Security: TCP/IP, Cisco IOS, VLANs, OSPF, ACLs, Zero Trust, Microsegmentation, PCI-DSS
Concepts: DevOps, Observability, Policy-as-Code, LLMs & RAG

EXPERIENCE

AriesTECH **Ottawa, ON**
Network Technician *Jan 2024 – Apr 2025*

- Installed, configured, and maintained Cisco routers & switches; designed Layer 2/3 segmentation (VLANs, trunking, inter-VLAN routing) and implemented static, RIP, and OSPF routing to optimize traffic flow and improve security across the LAN/WAN.
- Maintained 99.9% network uptime via proactive monitoring, rapid diagnostics (using Wireshark and Cisco IOS tools), and regular preventative maintenance (firmware upgrades, ACL reviews, configuration backups).
- Automated routine network configuration tasks and documentation updates using Python scripts, reducing manual configuration time by 30% and minimizing human error during deployment windows.

SELECTED PROJECTS

ZTAP ↗ | *Go, eBPF (CO-RE), WFP, etcd, gRPC, Prometheus, AWS/Azure/GCP, Kubernetes*

- Developed a cross-platform zero-trust engine in Go, enforcing kernel-level policies via Linux eBPF, Windows WFP, and macOS pf. Leveraged etcd and gRPC for real-time distributed policy synchronization and high-availability cluster coordination.
- Built a hybrid cloud sync and observability suite for AWS/Azure/GCP, featuring real-time flow monitoring and SHA-256 hash-chained logs for automated zero-trust enforcement and NIST SP 800-207 compliance reporting.

NetScope ↗ | *Rust, libpcap, BPF, Zero-Copy Parsing, WebAssembly*

- Developed a high-performance network protocol analyzer in Rust, leveraging libpcap and zero-copy memory management to capture and dissect Ethernet, IP, and transport layer traffic at gigabit speeds.
- Designed a real-time flow tracking engine with TCP state machine analysis, kernel-level BPF filtering, and a WebAssembly-based dashboard for live network visualization and anomaly detection.

personal-ai-cli ↗ | *Python, Local LLM (Ollama), RAG, CLI*

- Built a privacy-first RAG chatbot using LangChain, Ollama, and ChromaDB to enable secure, offline querying of personal documents via local HuggingFace embeddings.
- Optimized vector database updates with an incremental SHA256 ingestion pipeline, and designed an interactive Rich CLI featuring streaming LLM responses, memory, and source citations.

cloud-netmapper ↗ | *Go, AWS SDK, Network Visualization, Security Analysis, CLI*

- Built a Go CLI using AWS SDK v2 to automatically map VPC topologies across EC2, RDS, Lambda, and EKS, generating interactive visualizations and multi-format reports for architecture reviews.
- Developed a security analysis engine to detect network misconfigurations like open security groups and missing flow logs, featuring state-diffing to track infrastructure drift over time.